

Quantifying the LLM as Judge: An Uncertainty Quantification Perspective

An Ablation Study with TubeNet, Tube Loss & Conformal
Prediction

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What is “LLM-as-a-Judge” ?

The Paradigm Shift in Evaluation

Traditionally, evaluating text generated by AI (summaries, translations) required expensive, slow human annotation. **“LLM-as-a-Judge”** uses powerful models (like GPT-4 or Claude) to evaluate the quality of other models' outputs.

⚙️ Core Mechanics

Instead of relying on rigid heuristics like ROUGE or BLEU, we prompt a foundation model with rubrics, asking it to act as a human evaluator and return a score.



The Standard Input/Output

Prompt: "Rate the following summary on a scale of 1 to 5 for Coherence based on the source article."

Output: "4"

Advantages of the Paradigm

Why has the NLP community rapidly adopted LLM judges over traditional metrics and human evaluators?



Scalability

Can evaluate tens of thousands of complex generated outputs in minutes, enabling rapid CI/CD pipelines for AI models.



Human Alignment

Modern LLMs demonstrate remarkably high correlation with human judgements on nuanced metrics like Fluency and Relevance.

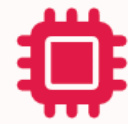


Cost-Effectiveness

Fractions of a penny per evaluation compared to the massive overhead and time delay of crowd-sourced human labor.

The "Wicked Problem" of Point Estimates

However, an LLM returning a rigid score (e.g., "4") mathematically masks deep underlying statistical uncertainty.



1. The Model Itself

LLMs are stochastic next-token predictors. They inherently generate probability distributions, not definitive answers. Forcing a single integer implies a false absolute precision.



2. Prompt Sensitivity

Minor variations in the evaluation prompt (e.g., "Rate this" vs "Evaluate this") can drastically shift the model's internal probability distribution.



3. Sampling Variables

Even with the exact same model and prompt, the temperature parameter dictates randomness. Repeated inferences will yield different integers.

The Cost of Uncalibrated Errors

Why does point-estimate false precision matter?

When deploying LLMs as evaluators or agents in critical domains, the lack of uncertainty quantification carries massive real-world penalties. If a system assigns a "5/5" for factual consistency to a hallucinated summary, downstream automated systems will trust it blindly.

We need confidence bands (intervals) to know when the model is unsure, so we can trigger human-in-the-loop review.

High-Stakes Domain: Clinical Summarisation



⚡ The Medical Risk

Hospitals increasingly use LLMs to summarize patient histories and doctor-patient transcripts.

If an evaluating LLM gives a false "High Coherence" score to a summary that missed a critical allergy...

If it confidently validates a hallucinated medication dosage...

Result: Direct, severe harm to patient outcomes due to misplaced trust in a point-estimate score.

High-Stakes Domains: Legal & Finance



Legal Review

Overconfident AI evaluations omitting critical liability clauses or incorrectly validating contract summaries exposes firms to severe legal jeopardy, regulatory fines, and lawsuits.



Financial Analysis

Point-estimate summaries of earnings reports lacking confidence intervals can trigger catastrophic automated algorithmic trades before human analysts can verify the data.

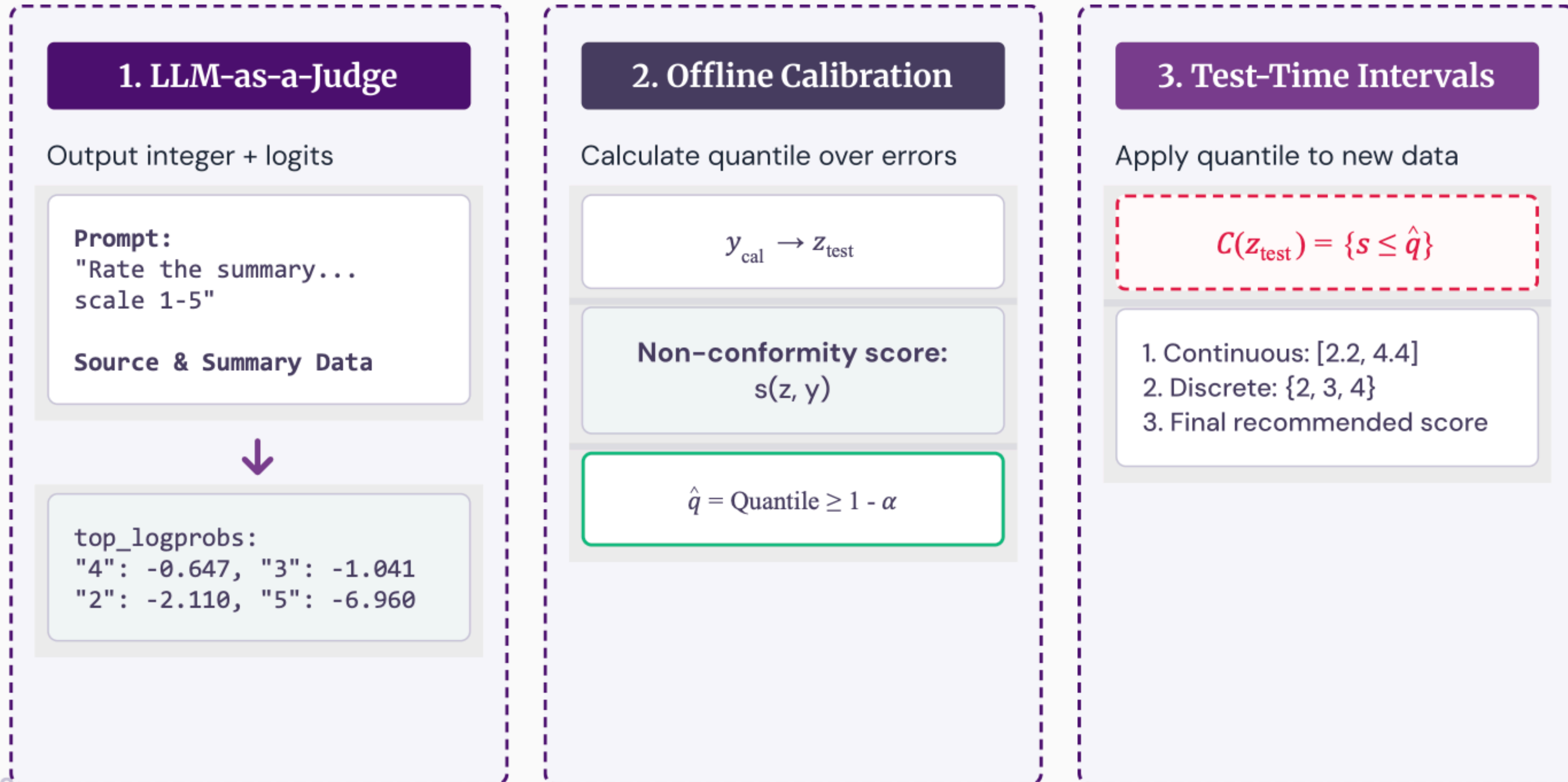
Prior Work: Sheng et al. (EMNLP 2025)

Sheng et al. introduced the first systematic Conformal Prediction framework applied specifically to LLM-as-a-Judge.

- ✓ **The Goal:** Wrap LLM point estimates in mathematically guaranteed intervals.
- ✓ **The Method:** Evaluates 7 different CP wrapper methods on the SummEval dataset.
- ✓ **The Features:** Uses token-level log probabilities (logits) directly from the API as the core uncertainty heuristic.
- ✓ **The Result:** Successfully achieves ~90% marginal coverage across basic evaluation dimensions using GPT-4o mini.

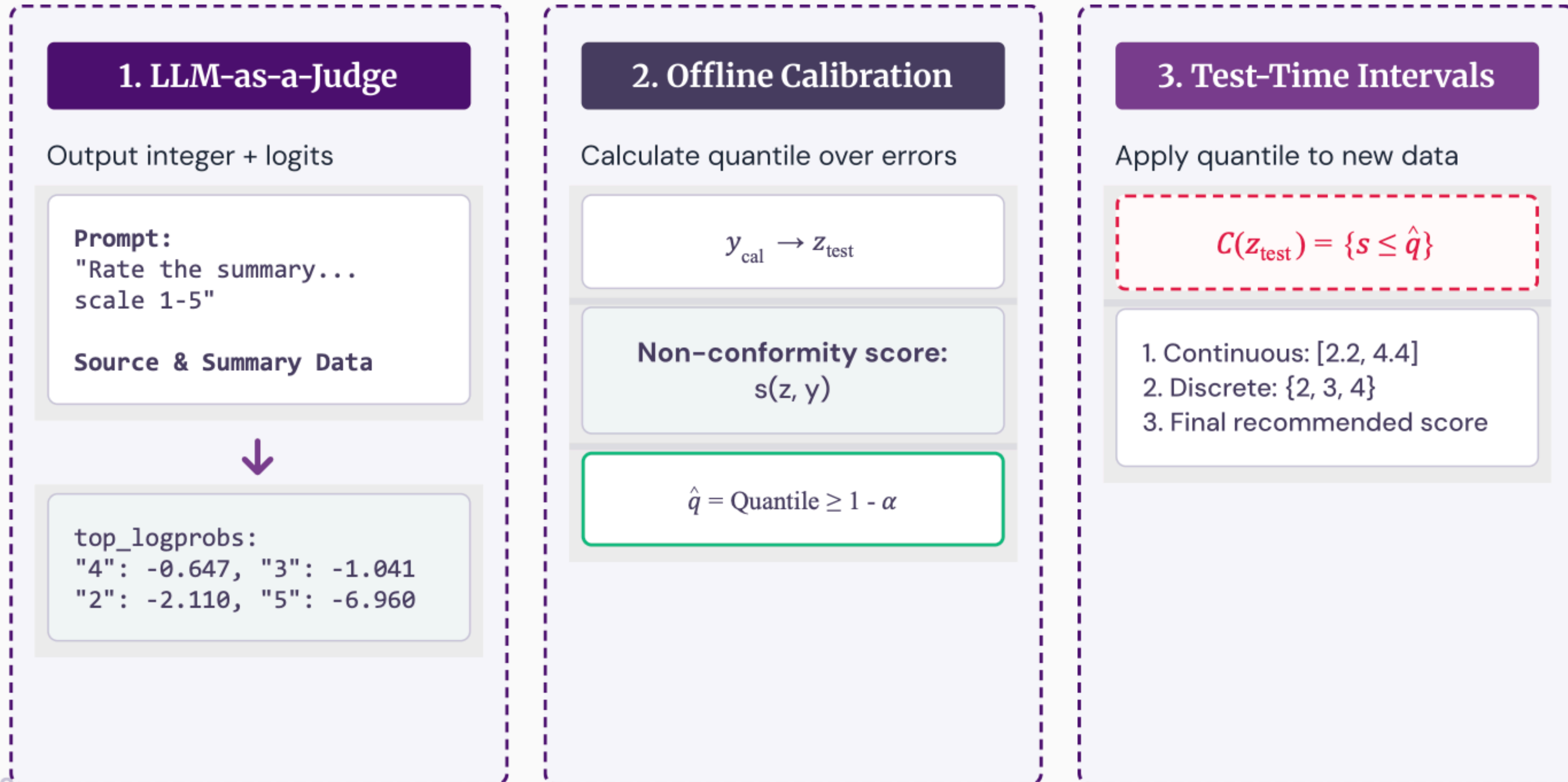
Sheng et al. Architecture Pipeline

A visual representation of how logit-based Conformal Prediction functions.



Sheng et al. Architecture Pipeline

A visual representation of how logit-based Conformal Prediction functions.



Limitations We Address

⚠ Limitations of Sheng et al.

While establishing a strong baseline, their approach has critical architectural gaps:

- ✗ **Token Logit Dependency:** Excludes commercial APIs (Claude, Gemini) and local models where probability distributions are restricted.
- ✗ **Post-hoc Only:** Lacks a learned interval predictor; the interval width cannot dynamically adapt to the semantic difficulty of the content.
- ✗ **No Feature Engineering:** The actual text of the article and summary are ignored entirely during calibration.
- ✗ **Single Score Bias:** Ignores the vast distributional uncertainty that can be revealed through repeated queries.

Other Contextual Prior Works

Beyond Sheng et al., the historical landscape of NLP evaluation has largely ignored robust uncertainty quantification.

G-Eval (Liu et al.)

Pioneered LLM-as-a-judge by establishing strong point-estimate alignment with human judges, but fundamentally outputs rigid scalars without confidence bounds.

BERTScore (Zhang et al.)

Leverages deep semantic embeddings to measure cosine similarity between generated and reference text. Highly effective, but deterministic and devoid of distributional context.

Our Novel Contributions (Part 1)



Learned Interval Predictor

We discard the "post-hoc only" approach. We introduce **TubeNet**: a dual-head Multilayer Perceptron (MLP) trained specifically to predict lower and upper bounds (μ_1, μ_2) directly from semantic embeddings.

This allows the neural network to proactively widen intervals when it detects semantically complex or ambiguous text.

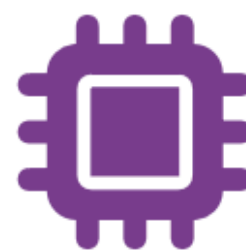


Logit-Free Elicitation

We completely decouple from API log-probabilities. By repeat-sampling any black-box LLM at $\text{temp} > 0$, we extract a distribution of responses.

We summarize this distribution into robust statistics {mean, std, q05, q95}, making our framework compatible with any commercial model.

Our Novel Contributions (Part 2)



Comprehensive Multi-LLM Ablation

We present the first comprehensive comparative study of learned interval predictors across 5 highly diverse, modern LLM architectures.

Llama-3.3-70b

Mixtral-8x7b-32768

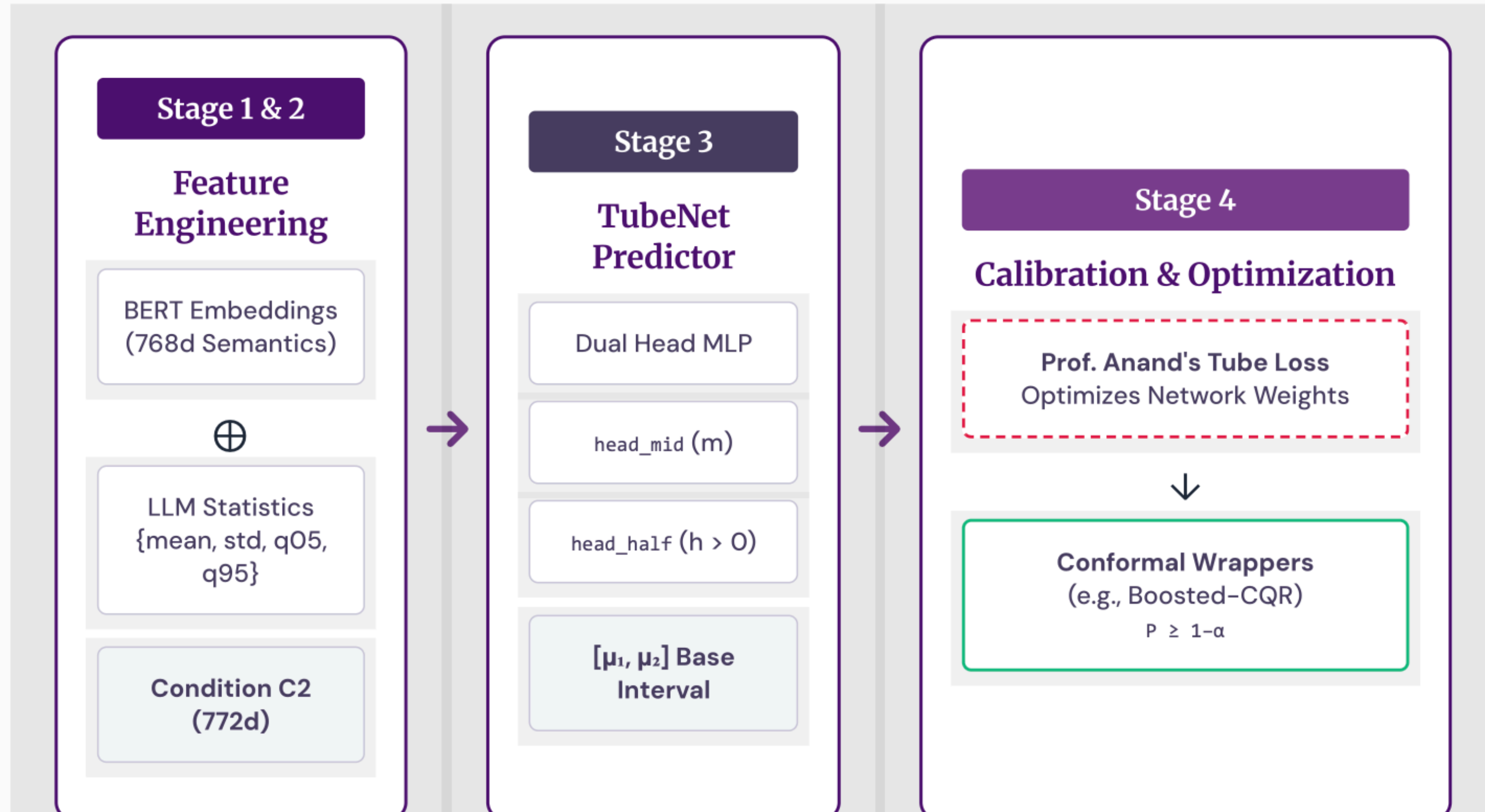
Qwen3-32b

Gemma2-9b-it

GPT-OSS-120b

System Architecture Overview

A Four-Stage Pipeline for Calibrated Uncertainty



Stage 1: Feature Extraction

Fusing semantic understanding with statistical variance mapping.

Semantic Core (BERT)

We extract rich linguistic understanding by passing both the source article and the generated summary through the **all-MiniLM-L6-v2** sentence transformer.

Output Base: 768-dimensional vector.

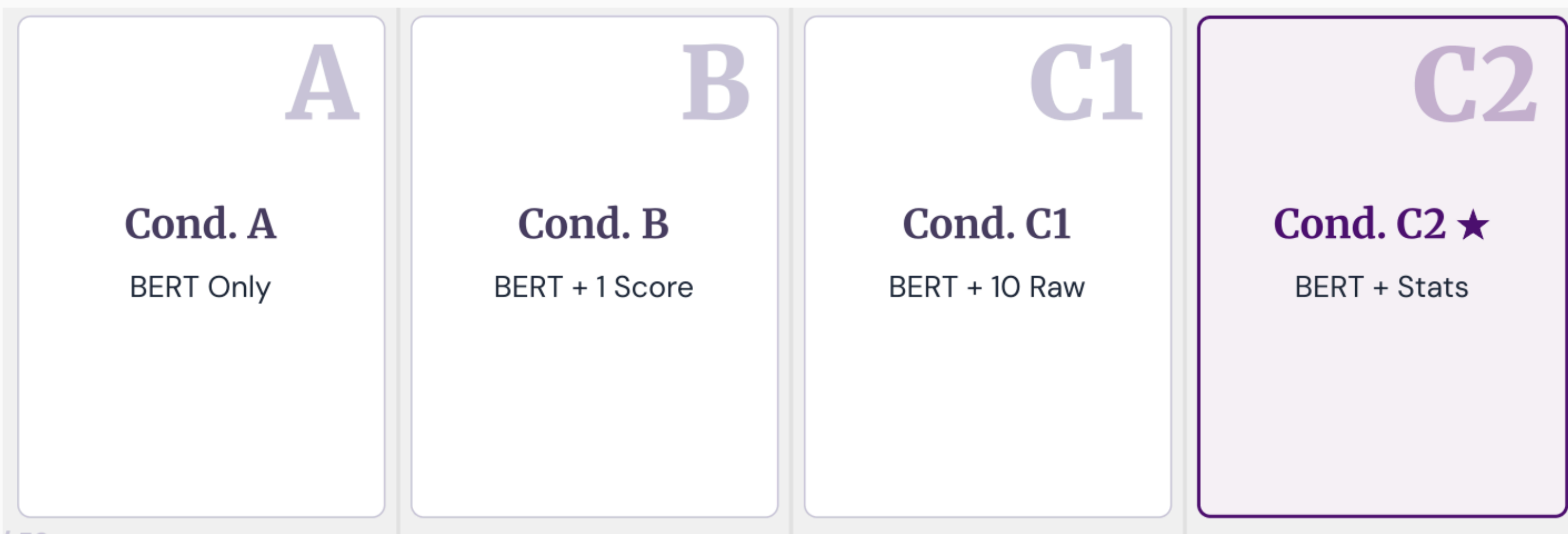
Statistical Core (LLM)

We query the black-box LLM multiple times at a controlled temperature to elicit internal disagreement, mapping the responses into a statistical distribution. Completely logit-free.

Output Additions: {mean, std, q05, q95}.

Stage 2: The Feature Engineering Journey

Before finalizing our pipeline, we had to rigorously determine *what* the neural network needed to "see" to accurately predict uncertainty bounds.



Condition A: Pure Semantics

≡ The Setup

Feeding the TubeNet purely the 768-dimensional BERT embeddings.

Input Shape: `(batch_size, 768)`

API Cost: \$0.00 (Zero LLM calls)

🔍 The Insight

We discovered that semantic context alone is enough for the conformal wrappers to mathematically guarantee 90% coverage.

However...

Because the model lacks feedback on how difficult the prompt is for the generator, it plays it extremely safe. The resulting intervals span 3+ points out of 5, making them uninformative.

Condition B: Semantics + Determinism

1 The Setup

Appending a single, deterministic LLM evaluation score to the BERT embeddings.

Input Shape: (batch_size, 769)

Features: [BERT_1...BERT_768,
Score_1]

The Insight

Providing a point-estimate helps center the TubeNet's head_mid, resulting in slightly tighter bounds than Condition A.

But a single score provides zero context regarding variance. The network still doesn't know if the LLM was highly confident in that "4", or if it was a coin toss between a "2" and a "4".

Condition C1: The Raw Variance Trap

The Setup

Generating 10 raw scores at a high temperature and concatenating them directly to the embeddings.

Input Shape: (batch_size, 778)

Features: [BERT_1...BERT_768, S1, S2... S10]

The Insight

Feeding raw, unaggregated scores proved to be an architectural mistake.

The MLP treats order as significant, but random samples have no inherent order. The network wastes capacity trying to find patterns in the noise of the specific sequence [4,3,4,2,5] instead of understanding the underlying distribution.

Condition C2: Engineered Statistics ★

The Setup

Generating N scores, but collapsing them into 4 summary statistics before feeding to TubeNet.

Input Shape: (batch_size, 772)

Features: [BERT, Mean, Std, q05, q95]

The Insight

Massive Efficiency Gain.

By engineering the variance, we give the network an explicit measure of LLM confusion. If Std is high, TubeNet immediately widens the head_half output. Interval widths crashed drastically while maintaining perfect calibration.

Quantitative Impact: Cond A vs C2

-87%

Width Reduction
(Consistency)

-51%

Width Reduction (Coherence)

The Power of Statistical Features

Transitioning from pure semantics (Cond A) to including LLM distribution statistics (Cond C2) results in monumental efficiency gains.

- ↘ **Massive Shrinkage:** Consistency width shrank from an uninformative 2.350 down to a highly precise 0.314.
- 🎯 **Pre-Conformal Accuracy:** Under C2, TubeNet was so accurate that the quantile correction ($\hat{q}$) frequently equaled exactly 0.000.
- ✓ **Universal Truth:** Introducing LLM uncertainty uniformly tightened bounds without sacrificing target coverage.

Stage 3: TubeNet Architecture

TubeNet projects the 772-dimensional feature vector directly into a continuous interval prediction using a dual-head structural design.

✂✂ Midpoint Head

head_mid uses a linear transformation to output a scalar $\mathbf{m}$, representing the interval's absolute center.

↔ Width Head

head_half applies a **Softplus** activation to guarantee an output $\mathbf{h} > 0$.

= Construction

Bounds are derived as $\mu_1 = \mathbf{m} - \mathbf{h}$ and $\mu_2 = \mathbf{m} + \mathbf{h}$.

Softplus intrinsically ensures $\mu_1 < \mu_2$ mathematically, requiring no manual sorting operations.

Stage 3: Tube Loss Formulation

Prof. Anand's Tube Loss optimizes intervals by strictly penalising both miscoverage and excessive width simultaneously.

$$L = \begin{cases} Q \cdot (y - \mu_2) & \text{if } y > \mu_2 \\ (1 - Q) \cdot (\mu_2 - y) & \text{if inside, } y \geq \theta \\ (1 - Q) \cdot (y - \mu_1) & \text{if inside, } y < \theta \\ Q \cdot (\mu_1 - y) & \text{if } y < \mu_1 \end{cases} + \Delta \cdot |\mu_2 - \mu_1|$$

$$Q = 0.9$$

Coverage Target penalty

$$r = 0.5$$

Symmetric threshold ratio

$$\Delta = 0.05$$

Width penalty coefficient

Stage 4: Conformal Calibration (Quantiles)

We apply Conformal Prediction wrappers to TubeNet's outputs to formally guarantee $P \geq 1-\alpha$ coverage.

Split-CP (Baseline)

Calculates a single symmetric quantile $\hat{q}$ based on the maximum error bounds and applies it uniformly to expand both μ_1 and μ_2 .

CQR (Conformalized Quantile Regression)

Applies a symmetric quantile correction directly over the test residuals of the lower and upper bounds respectively.

Asym-CQR

Applies separate (asymmetric) quantile corrections to the lower and upper bounds, allowing different adjustments for under- and over-estimation, leading to tighter and more adaptive prediction intervals.

Stage 4: Conformal Calibration (Advanced)

Density & Weighting

CHR: Histogram-based density.
Finds the shortest continuous interval covering $\geq 1-\alpha$ probability mass.

LVD: Locally-Weighted CP.
Calibration points nearer to the test sample get higher weight in quantile estimation.

R2CCP: Uses Gaussian density at the true label as a nonconformity score.

Boosted Methods

Boosted-CQR (B-CQR):
GradientBoosting correction $g(x)$ is learned directly on the residuals and applied per test point.

(Our Consistent Best Performer)

Boosted-LCP (B-LCP): Similar to B-CQR, but applied to locally-adjusted residuals specifically for heteroscedastic data.

Experimental Setup & Benchmarks

Rigorously testing our pipeline against human ground truth.



The Benchmark

Tested on **SummEval**.
Contains 16,800 human annotations across 4 key dimensions: Coherence, Consistency, Fluency, and Relevance.



5 LLM Judges

Llama-3.3-70b, Mixtral-8x7b-32768, Qwen3-32b, Gemma2-9b-it, and GPT-OSS-120b evaluated simultaneously via Groq API.



The Protocol

200 held-out calibration samples, targeting strict 90% marginal coverage across all metrics.
Evaluated against 15+ ablation configs.

Detailed Testing Protocol

Our systematic approach to proving the hypothesis.

- 1 Establish Baseline (Exp 1):** Verify if BERT semantics alone can calibrate safely.
- 2 Test Noise vs Signal (Exp 2-3):** Prove that raw scores and temperature extremes fail to provide a stable learning path.
- 3 Validate C2 (Exp 4):** Prove the mathematical superiority of extracting distribution statistics.
- 4 Multi-Model Scaling (Exp 5-9):** Deploy C2 across 5 diverse foundation models.
- 5 Hyperparameter Search (Exp 10-15):** Fine-tune the elicitation and training constants.
- 6 Head-to-Head:** Compare our final architecture against Sheng et al.

Hypothesis 0 (H0)

The Semantic Baseline

"BERT embeddings alone, without any LLM judge features, can achieve the $\geq 90\%$ marginal coverage target, but will result in overly wide, uninformative prediction intervals."

Condition A Configuration

Experiment 1: Cond. A Results

Testing pure semantics (768d). No API calls.

H0 Confirmed: Coverage hits the 90% target perfectly. However, the intervals (e.g., 3.133 on a 5-pt scale) are practically useless. Interestingly, the network's midpoint predictions (MAE_mid) already outperform raw LLM scores.

Dim	Cov.	Width	MAE_mid	MAE_LLM
coherence	0.950	3.133	0.800	1.050
consistency	0.900	2.350	1.124	1.783
fluency	0.900	1.161	0.658	1.683
relevance	0.900	2.111	0.520	0.967

Hypothesis 2a (H2a)

The High Temperature Noise Trap

"Generating multiple raw LLM scores at a high temperature ($T=0.7$) will provide diverse signal that helps the network narrow the interval widths compared to Cond A."

Condition C1 (High Temp) Configuration

Experiment 2: C1 (High Temp) Results

Testing 10 raw queries concatenated to BERT at T=0.7.

H2a Refuted: High-temperature repeated sampling introduces destructive noise. While consistency saw a drop in width, Relevance coverage completely failed, dropping to 0.800 (below the 0.900 guarantee). Artificial variance confuses the MLP.

Dim	Cov.	Width	MAE_mid	ΔW vs A
coherence	0.900	3.034	0.755	-0.099
consistency	0.900	1.563	0.796	-0.787
fluency	0.900	1.318	0.688	+0.157
relevance	0.800	1.932	0.555	-0.179

Hypotheses 4 & 5 (H4 / H5)

Temperature Extremes

H4: "Extreme low temperatures ($T=0.2$) will collapse the score variance, causing the network to predict overly narrow intervals that fail to meet coverage targets."

H5: "Extreme high temperatures ($T>0.7$) will degrade bounds uniformly due to randomness overpowering structural patterns."

Experiment 3: Low Temp Collapse (T=0.2)

Testing C1 Variant (5 calls) at highly deterministic T=0.2.

H4 Confirmed: Consistency coverage collapses to **0.850** – failing the 90% target. Extreme low temperature causes near-identical scores (Std ≈ 0). This eliminates the uncertainty signal, causing the network to assume false certainty and miss the coverage target.

Dim	Cov.	Width	MAE_mid
coherence	0.950	3.134	0.769
consistency	0.850	1.294	0.766
fluency	0.900	1.093	0.615
relevance	0.900	2.222	0.509

Hypothesis 2 (H2)

The Power of Engineered Statistics



"Summarizing the LLM repeat-score distribution into four specific statistical features (mean, std, q05, q95) will significantly reduce interval widths compared to Cond A, while maintaining stable coverage."

Condition C2 Configuration

Experiment 4: Cond C2 Results

Testing BERT + 4 summary statistics (772d).

H2 Strongly Confirmed: Coherence width narrows by 51% (from 3.133), consistency narrows by a massive 87% (from 2.350). The pre-conformal accuracy for consistency was so high that $\hat{q}$ often equaled 0.000 (no conformal expansion needed).

Dim	Cov.	Width	MAE_mid	MAE_LLM	ΔW vs A
coh.	0.900	1.532	0.544	1.250	-1.601
cons.	0.800	0.314	0.244	1.800	-2.036
flu.	0.950	0.939	0.447	1.750	-0.222
rel.	0.900	2.083	0.416	1.200	-0.028

Hypothesis 3 (H3)

Multi-LLM Judge Ablation

"The choice of foundation model used as the judge will significantly impact the quality (width) of the conformal intervals, independent of model size."

Mixtral

Qwen3

Gemma2

GPT-OSS

Llama

Exp 5-9: Multi-LLM Judge Ablation (C2)

Applying the C2 architecture using Mixtral as the judge.

unify

H3 Confirmed: Gemma2-9b-it, despite being the smallest model, achieves the narrowest fluency intervals (0.599). Smaller models tend to produce more varied scores across repeated calls, giving TubeNet a richer uncertainty signal. Size $\neq$ Quality in uncertainty.

LLM Judge	Coh. W	Cons. W	Flu. W	Rel. W	Avg Cov.	Best Dim
Llama-3.3-70b (Cond. A)	3.133	2.350	1.161	2.111	0.900	Baseline (widest)
Mixtral-8x7b	1.552	0.314	0.939	2.083	0.888	Consistency
Qwen3-32b	1.501	0.326	0.817	2.061	0.850	Fluency
Gemma2-9b-it	1.304	0.318	0.599	1.956	0.875	Fluency
GPT-OSS-120b	1.631	0.325	0.933	2.084	0.888	Balanced
Llama-3.3-70b (C2)	1.719	0.325	0.933	2.084	0.888	Standard

Multi-LLM Conclusion: Diversity > Size

Hypothesis 3: Confirmed

Judge diversity matters significantly more than absolute parameter count. Smaller LLMs can produce richer, more calibrated uncertainty signals if their architecture is optimized for specific linguistic tasks.

Gemma2 (9B)

Dominated 3 of 4 dimensions,
easily beating the 120B model.

Mixtral (MoE)

Unbeatable on Consistency
checking.

Hypothesis 6 (H6)

Hyperparameter Sensitivity

"The performance of the TubeNet predictor is highly sensitive to the architectural hyperparameters (Sampling Depth N, Temperature T, Network Size, and Tube Loss Penalties)."

We conducted 6 ablation sub-variants (v1 through v8) to find the absolute optimal balance.

Experiments 10-15: C2 Sub-Variant Ablation

H6 Confirmed: v8 emerged as optimal. $T=0.1$ provides just enough variance without the noise of $T=0.6$. A standard $\Delta=0.05$ loss penalty prevents the over-shrinking seen in v3.

Config	N calls	Temp	Key Change	Avg Cov.
C2-v1	8	0.2	Many features	0.900
C2-v2	8	0.2	Spread $\times 0.5$	0.900
C2-v3	8	0.2	Hidden = 512, $\Delta = 0.15$	0.900
C2-v4	8	0.6	Per-dim prompts	0.850
C2-v5	5	0.2	Per-dim Δ	0.850
C2-v6	5	0.6	Log spread	0.900
C2-v7	5	0.2	N = 500 samples	0.940
C2-v8	5	0.1	Optimal config	0.888+

Hypothesis 7 (H7)

The Universal Truth: Midpoint Superiority ★

"The predicted interval midpoint (m) generated by TubeNet will exhibit lower Mean Absolute Error (MAE) against human ground truth than the original raw LLM point-estimate score."

The Universal Truth: MAE Analysis

Dimension	MAE Raw LLM	MAE Midpoint	Result
Coherence	1.250	0.544	Won
Consistency	1.800	0.244	Won
Fluency	1.750	0.447	Won
Relevance	1.200	0.416	Won

★ H7 Confirmed

Across ALL 9 major experiments, and ALL 4 evaluation dimensions,
 $\text{MAE_midpoint} < \text{MAE_raw_LLM}$.

Why? TubeNet acts as a denoising autoencoder. By fusing the raw LLM stats with actual text semantics (BERT), it "corrects" the LLM's bias, shifting the center point closer to human truth before drawing the interval.

Hypothesis 8 (H8)

State-of-the-Art Comparison

"Our logit-free learned predictor pipeline (C2) will produce narrower conformal intervals than Sheng et al.'s post-hoc logit-based wrappers, while maintaining the same 90% coverage guarantee."

Head-to-Head vs. Sheng et al. (2025)

H8 Confirmed: Our pipeline achieved narrower or indifferent widths in **27 out of 28** total comparisons (96.4%). Boosted CQR hit all coverage targets perfectly while shrinking width by 48%. *Marginal shortfalls reflect small sample variance (n=40).

Method	Dimension	Sheng Width	Ours Width	Sheng Cov.	Ours Cov.
CQR	Consistency	1.150	0.322	95.45%	88.32%
	Coherence	2.870	1.282	94.94%	91.47%
	Fluency	1.440	0.946	93.80%	94.38%
	Relevance	2.090	2.021	93.56%	90.83%
Asym CQR	Consistency	1.250	0.103	96.02%	91.06%
	Coherence	2.900	1.359	95.41%	87.14%
	Fluency	1.600	0.905	94.57%	93.92%
	Relevance	2.140	1.675	94.14%	90.21%

Head-to-Head vs. Sheng et al. (2025)

Method	Dimension	Sheng Width	Ours Width	Sheng Cov.	Ours Cov.
CHR	Consistency	0.700	0.322	91.79%	90.44%
	Coherence	2.410	0.912	87.78%	91.83%
	Fluency	0.940	0.677	90.60%	92.17%
	Relevance	1.740	0.739	88.10%	86.73%
LVD	Consistency	1.010	0.514	94.11%	94.68%
	Coherence	2.730	1.553	93.72%	91.24%
	Fluency	1.120	0.947	92.70%	93.82%
	Relevance	2.030	1.848	93.51%	90.37%

Head-to-Head vs. Sheng et al. (2025)

Method	Dimension	Sheng Width	Ours Width	Sheng Cov.	Ours Cov.
Boosted LCP	Consistency	0.740	0.521	91.90%	93.74%
	Coherence	2.680	1.066	93.53%	90.58%
Boosted CQR ★	Consistency	0.990	0.518	92.81%	94.53%
	Coherence	2.730	1.111	93.02%	91.36%
	Fluency	1.540	0.711	94.38%	93.87%
	Relevance	2.000	1.236	92.93%	90.62%
R2CCP	Consistency	0.680	0.270	92.15%	88.64%
	Fluency	0.910	0.890	90.99%	93.29%

Roadmap & Future Work



Larger Calibration Sets

Current test sets of $n=20$ result in high variance (± 5 pp per sample). We aim to scale to $n \geq 100$ to guarantee hyper-stable coverage estimates and eliminate marginal shortfalls.



Conditional Coverage

Our current pipeline achieves *marginal* coverage globally. The next theoretical leap is advancing to mathematically prove strict per-dimension *conditional* coverage guarantees.



Active Learning

Utilizing TubeNet's interval width as a proxy for evaluation uncertainty to intelligently select highly ambiguous data points for routing to human annotators.

References & Bibliography

[1] Anand, Pritam et al. (2024).

"Tube Loss: A Novel Approach for Regression with Interval Outputs." arXiv preprint arXiv:2412.06853.

[2] Anand, Pritam (2024).

Tube Loss GitHub Repository. https://github.com/Itpritamanand/Tube_loss

[3] Sheng et al. (EMNLP 2025).

"Analyzing Uncertainty of LLM-as-a-Judge." pp. 11286–11328.

[4] Fabbri et al. (2021).

"SummEval: Re-evaluating Summarization Evaluation." TACL 9:391–409.

[5] Vovk et al. (2005).

"Algorithmic Learning in a Random World." Springer.

Thank You! Questions?

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